Kinvecta

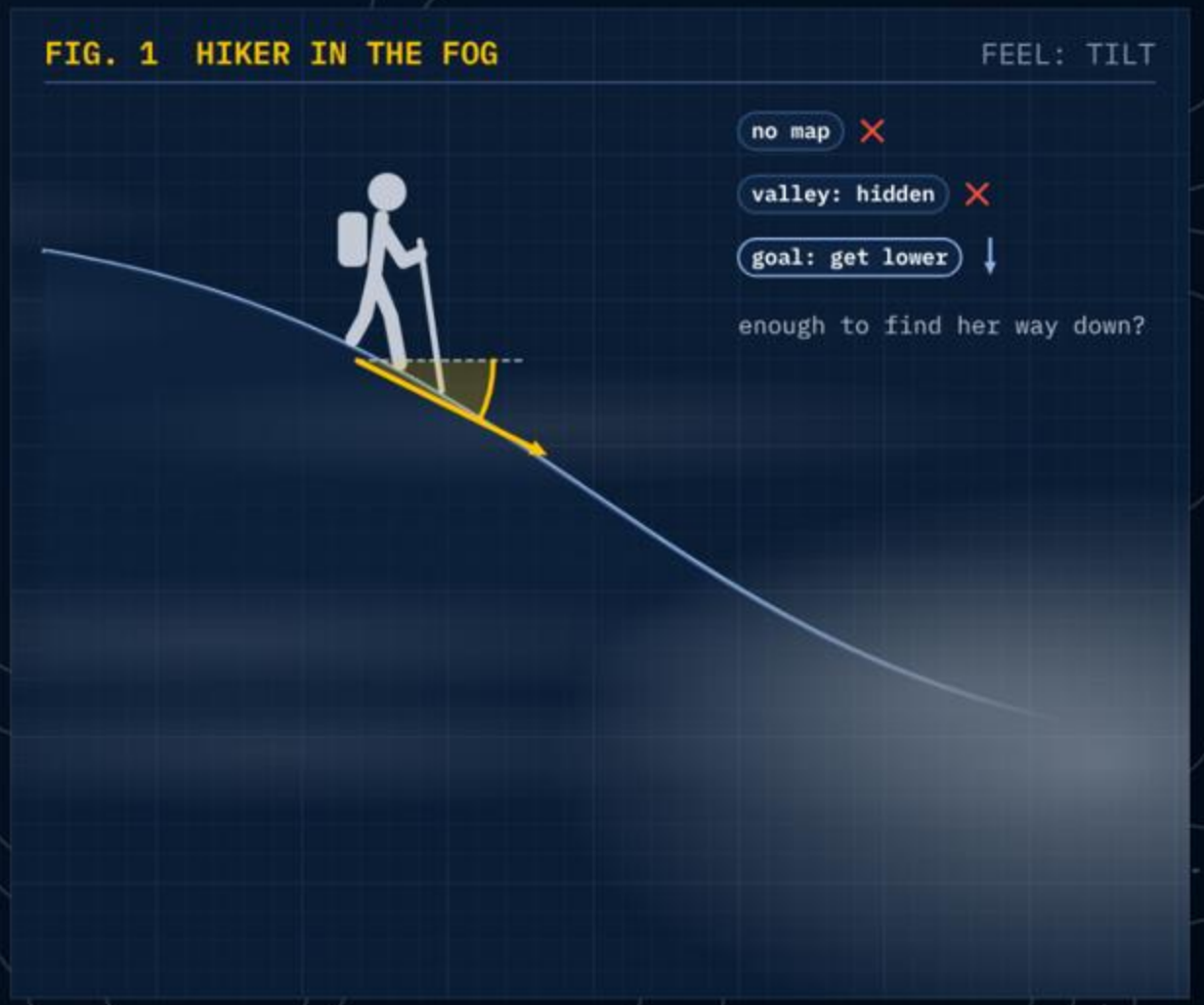
HOW LARGE LANGUAGE MODELS WORK · EPISODE 5

Lost in the fog. Feel the tilt.

No map, no view, just the slope under your boots. That one feeling is how a model learns.

13-minute lesson

Companion notebook



MARK

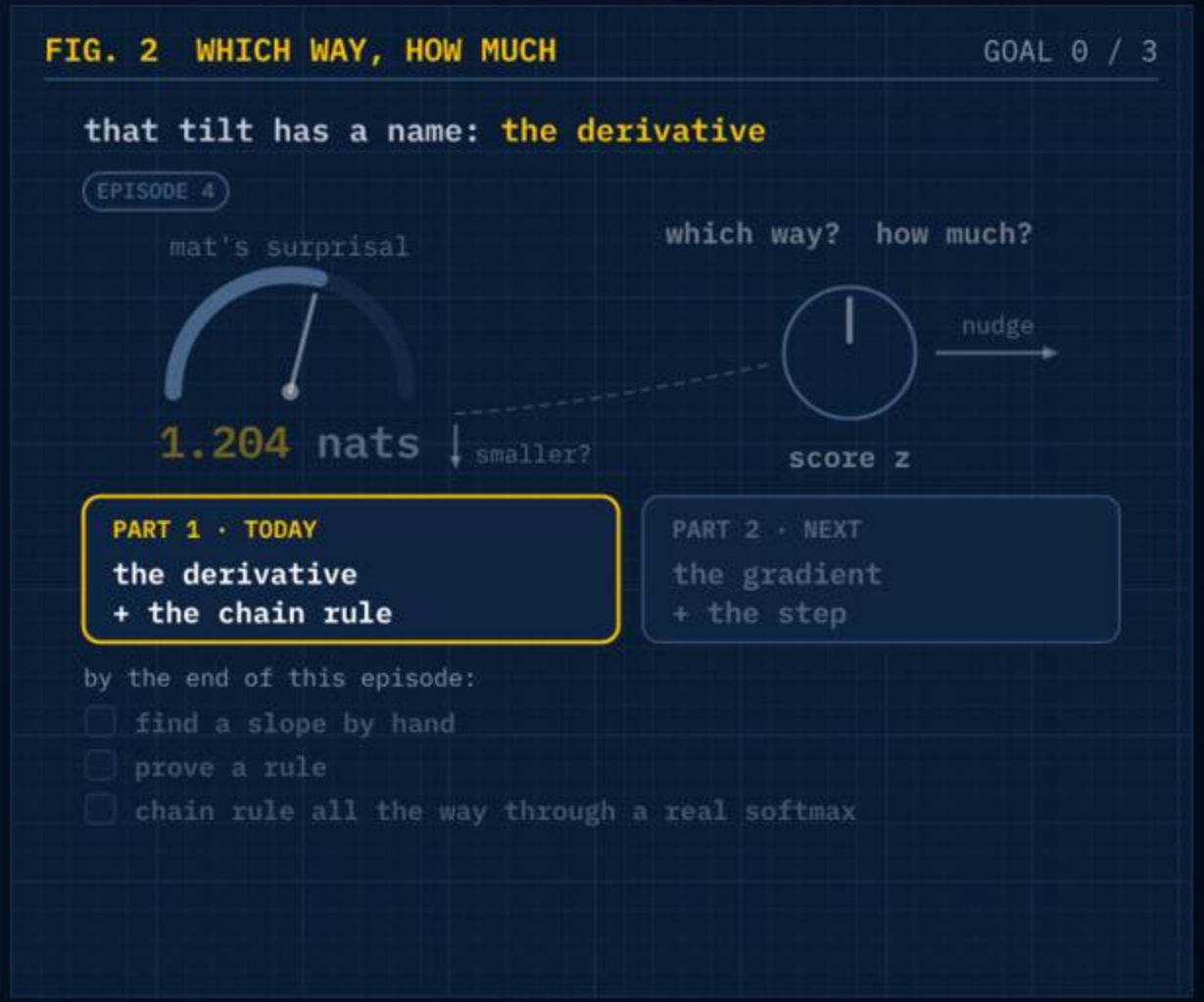
WHY THIS MATTERS

Which way, and how much?

Episode 4: mat's surprisal was 1.204 nats. Which way should we nudge the model to shrink it, and by how much?

01 Part 1 (today): the derivative and the chain rule

02 Part 2 (next): the gradient and the step



Surprisal −ln p from Episode 4. Episode 1's probabilities (mat 0.30, floor 0.20, sofa 0.15, other 0.35) are illustrative, not measured.

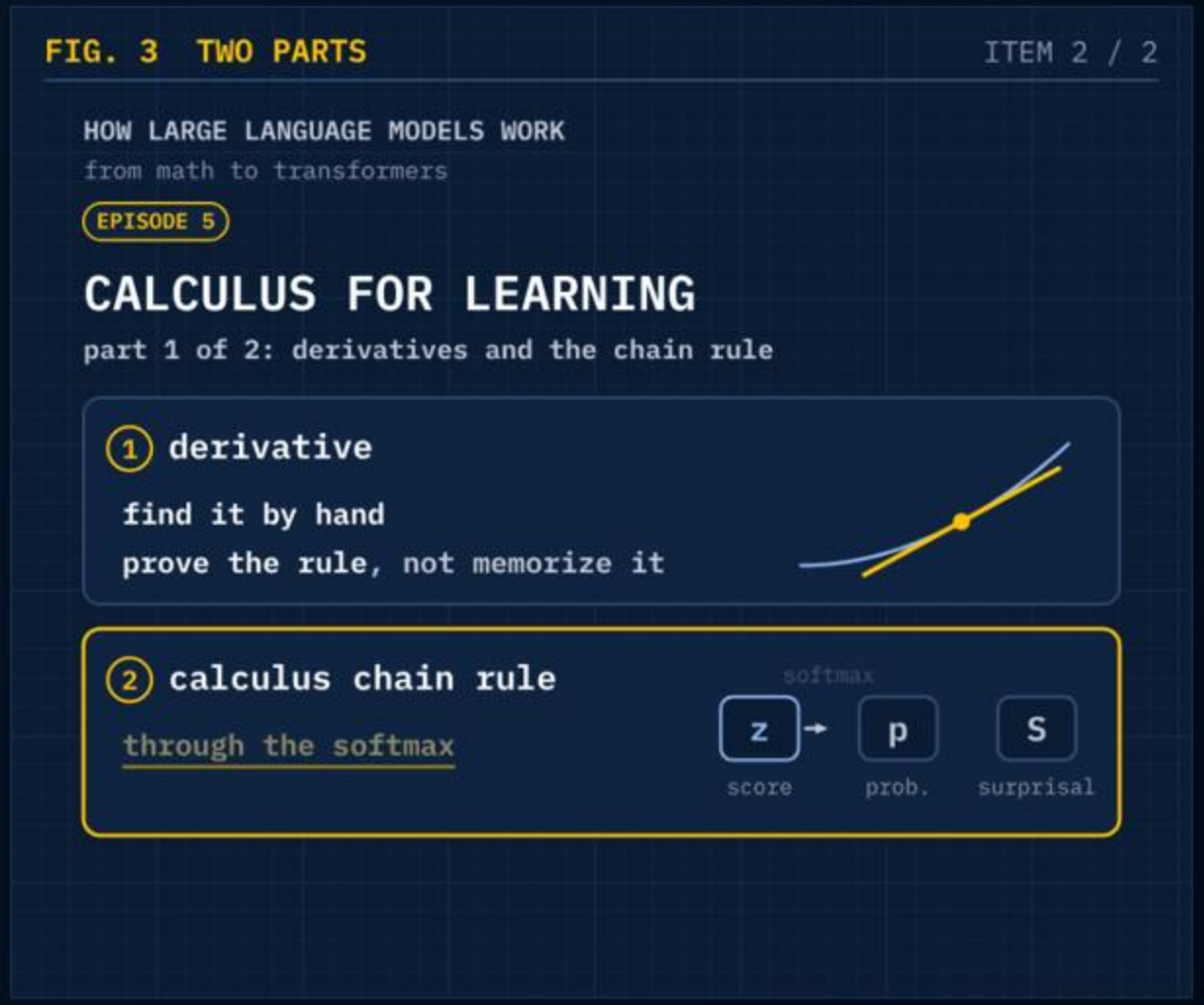
EPISODE 5 · CALCULUS FOR LEARNING, PART 1

Derivatives and the chain rule

Two parts, every step written out, and a notebook with every number.

01 The derivative: find it by hand, prove the rule

02 The calculus chain rule, all the way through softmax



Companion notebook: "Episode 05 - Calculus for Learning" (Colab), linked in the description. Sources on every slide.

PART 1 · THE IDEA

A rate of change, right now

A speedometer reading 60 mph doesn't mean you'll drive 60 miles. It means: at this instant, distance grows at 60 per hour.

01 Derivative: how fast the output changes, per unit of input, at one point



Analogy for intuition; the definition follows on the next slide.

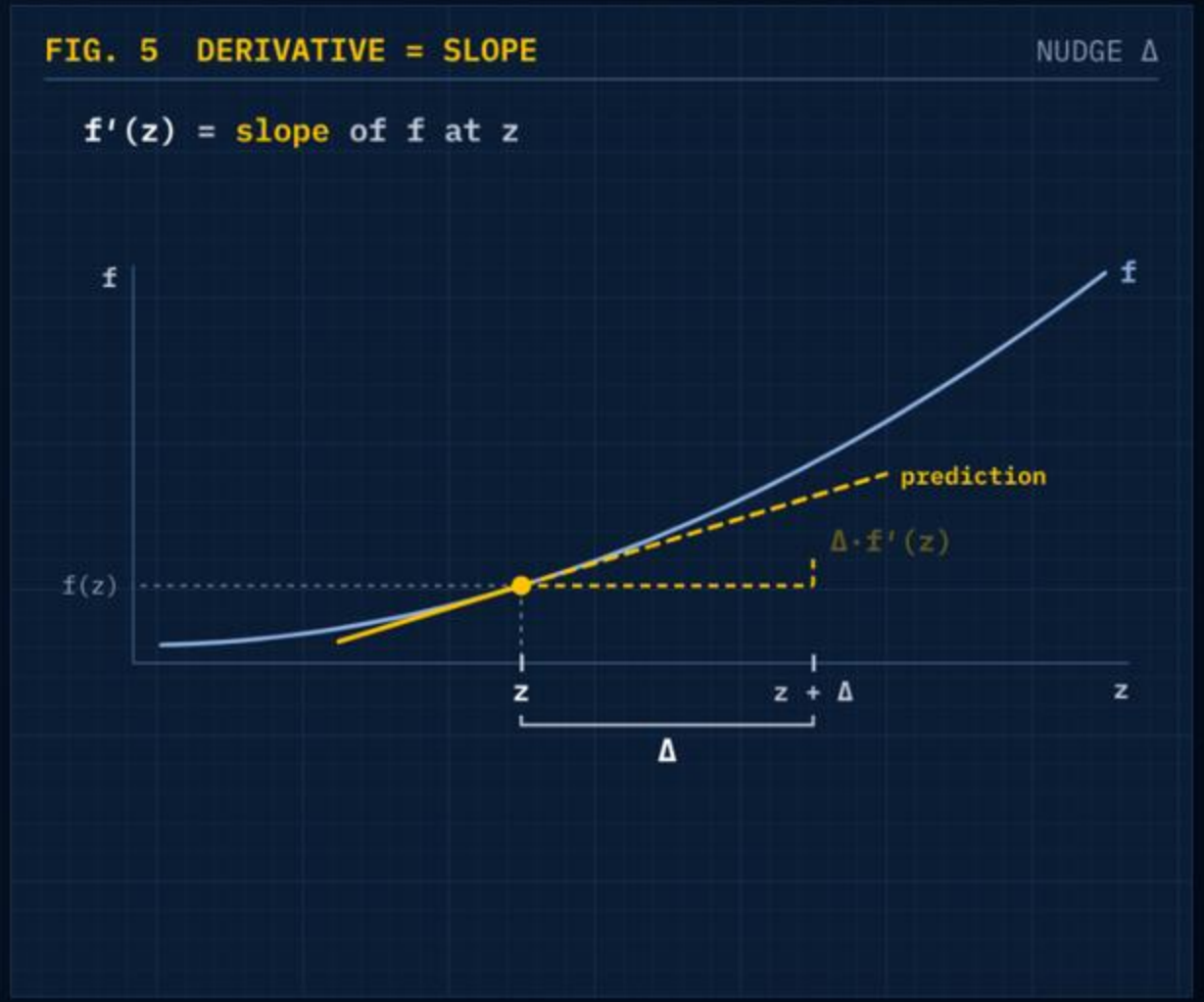
A derivative is a slope

$$f(z + \Delta) \approx f(z) + \Delta \cdot f'(z)$$

01 f'(z): how fast the output changes as z moves

02 Nudge the input by Δ

03 The output moves by about Δ × the slope



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 4 §4.3 (gradient-based optimization).

PART 1 · BY HAND

Find a slope the slow way

$f(z) = z^2$ at $z = 3$, so $f(3) = 9$

01 Step 1: $(16 - 9) \div 1 = 7$

02 Step 0.1: $(9.61 - 9) \div 0.1 = 6.1$

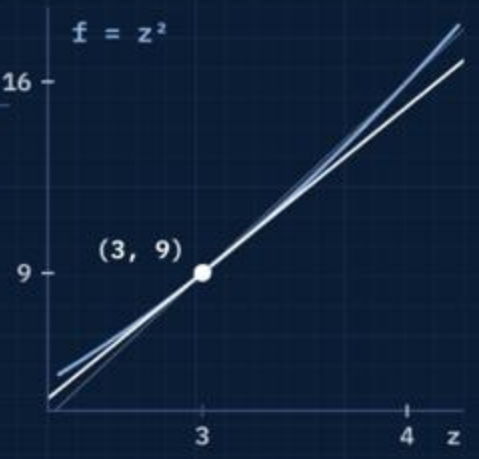
03 Step 0.01: $(9.0601 - 9) \div 0.01 = 6.01$

FIG. 6 SLOPE BY HAND — STEP 0.01

$f(z) = z^2$ at $z = 3$

$f(3) = 3^2 = 9$ by hand, the slow way

step 1, $z = 4$	f	$4^2 = 16$
	change	$16 - 9 = 7$
	slope	$7 \div 1 = 7$
step 0.1, $z = 3.1$	f	$3.1^2 = 9.61$
	change	$9.61 - 9 = 0.61$
	slope	$0.61 \div 0.1 = 6.1$
step 0.01, $z = 3.01$	f	$3.01^2 = 9.0601$



slopes

7

→ 6.1

→ ?

Worked example; every number is checked in the companion notebook.

MARK

PART 1 · PROVE IT BY HAND

Why z^2 has slope 2z

$$(z + \Delta)^2 = z^2 + 2z\Delta + \Delta^2$$

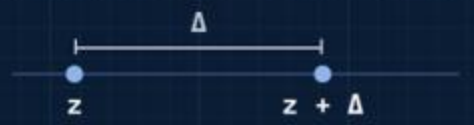
01 Change: $2z\Delta + \Delta^2$

02 Divide by Δ: $2z + \Delta$

03 Shrink Δ to 0: $2z$. At $z = 3$: 6

FIG. 7 PROVE IT: $z^2 \to 2z$ — slope = 2z

why is the slope of z^2 exactly 2z? BY HAND 3 LINES



step

new $(z + \Delta)^2 = z^2 + 2z\Delta + \Delta^2$

change $(z + \Delta)^2 - z^2 = 2z\Delta + \Delta^2$

÷ Δ $(2z\Delta + \Delta^2) \div \Delta = 2z + \Delta$

Δ → 0 $2z + \cancel{\Delta} \to \boxed{2z}$

z = 3 $2 \cdot 3 = \boxed{6}$

OpenStax, *Calculus Volume 1*, ch. 3 (differentiation rules, chain rule). Worked example; every number is checked in the companion notebook.

PART 1 · TWO RULES

Rules save the work

Power rule and the log rule.

01 $d/dz\, z^2 = 2z \rightarrow$ at 3: 6 ✓

02 $d/dz \ln z = 1/z$

FIG. 8 TWO RULES — RULE 2 / 2

by hand: 7 → 6.1 → 6.01 → 6

shortcut: rules

POWER RULE

$d/dz\ z^2 = 2z$

at z = 3 → 2·3 = 6 ✓



NATURAL LOG

the one we need for surprisal

OpenStax, *Calculus Volume 1*, ch. 3 (differentiation rules, chain rule).

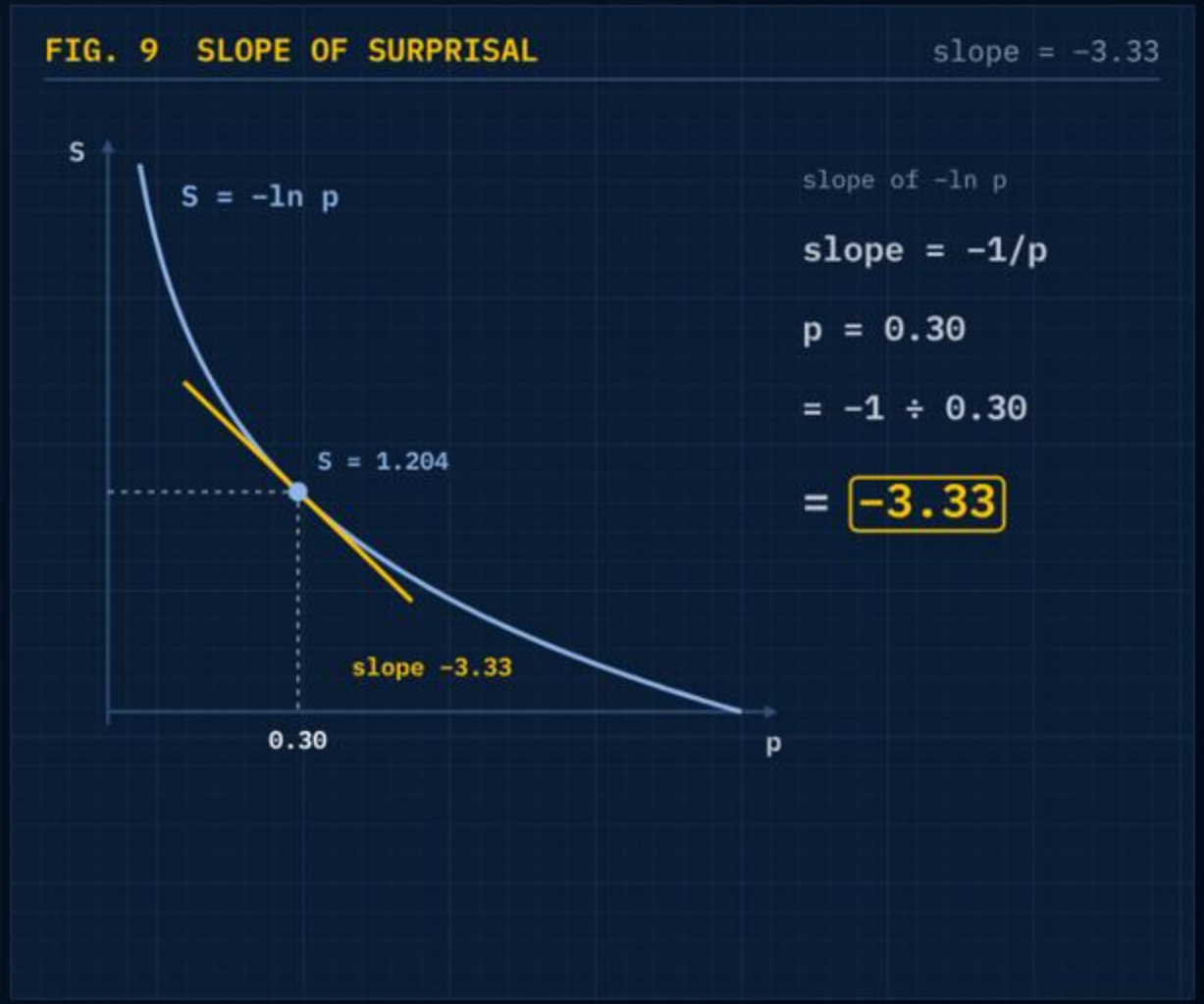
PART 1 · SURPRISAL'S SLOPE

The slope of −ln p

$S(p) = -\ln p$, so $S'(p) = -1/p$

01 At p = 0.30: −1 ÷ 0.30 = −3.33

02 Raise p a little, S falls ~3.33× as fast



OpenStax, *Calculus Volume 1*, ch. 3 (differentiation rules, chain rule). Episode 1's probabilities (mat 0.30, floor 0.20, sofa 0.15, other 0.35) are illustrative, not measured.

PART 1 · CHECK IT

Predict, then measure

Nudge p from 0.30 to 0.32 (Δ = 0.02).

01 Predict: 1.204 − 0.02 × 3.33 ≈ 1.137

02 True: −ln 0.32 = 1.139

FIG. 10 CHECK THE SLOPE ≈ 1.137

p: 0.30 → 0.32

Δ = 0.02

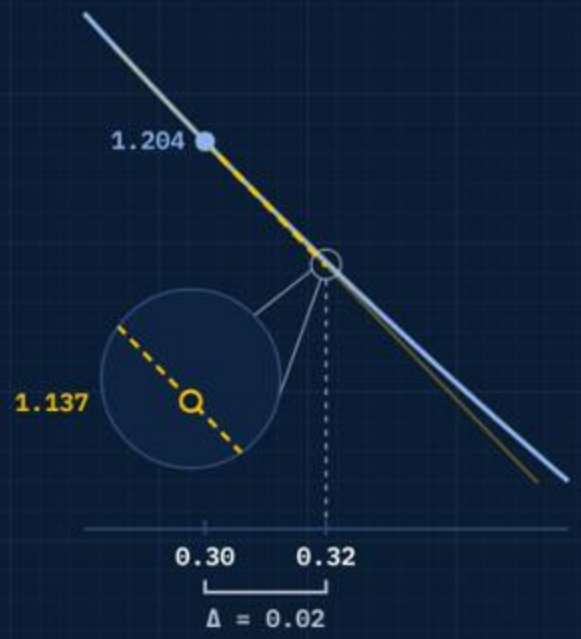
prediction

1.204 − 0.02 × 3.33

≈ 1.137

true value

−ln 0.32



Worked example; every number is checked in the companion notebook.

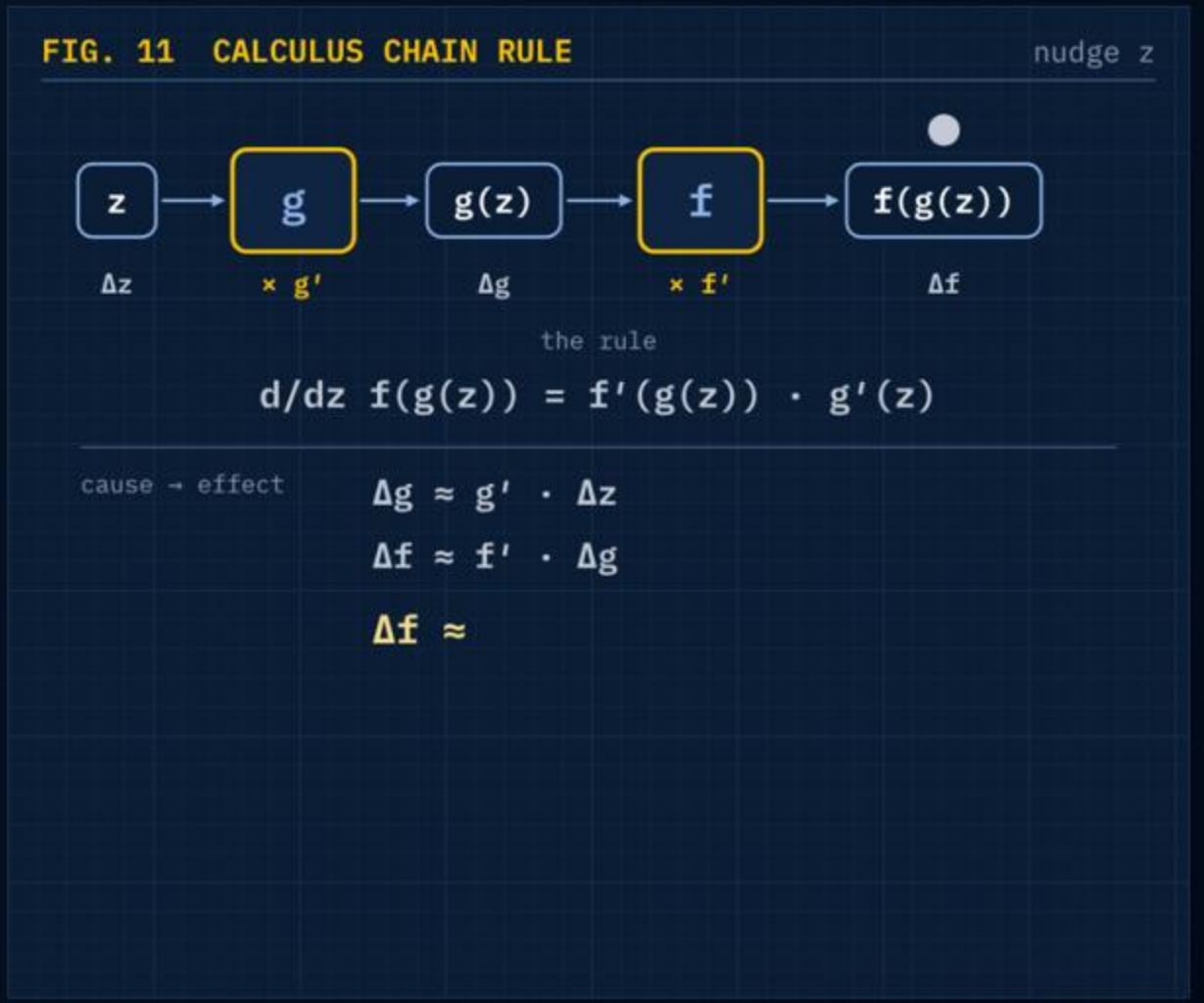
PART 2 · THE CALCULUS CHAIN RULE

Multiply the effects

$$d/dz\ f(g(z)) = f'(g(z)) \cdot g'(z)$$

01 Nudge z: g moves by g' × the nudge

02 That moves f by f' × the move in g



3Blue1Brown, "Visualizing the chain rule and product rule," *Essence of calculus*, ch. 5 (2017). OpenStax, *Calculus Volume 1*, ch. 3 (differentiation rules, chain rule).

PART 2 · BY HAND

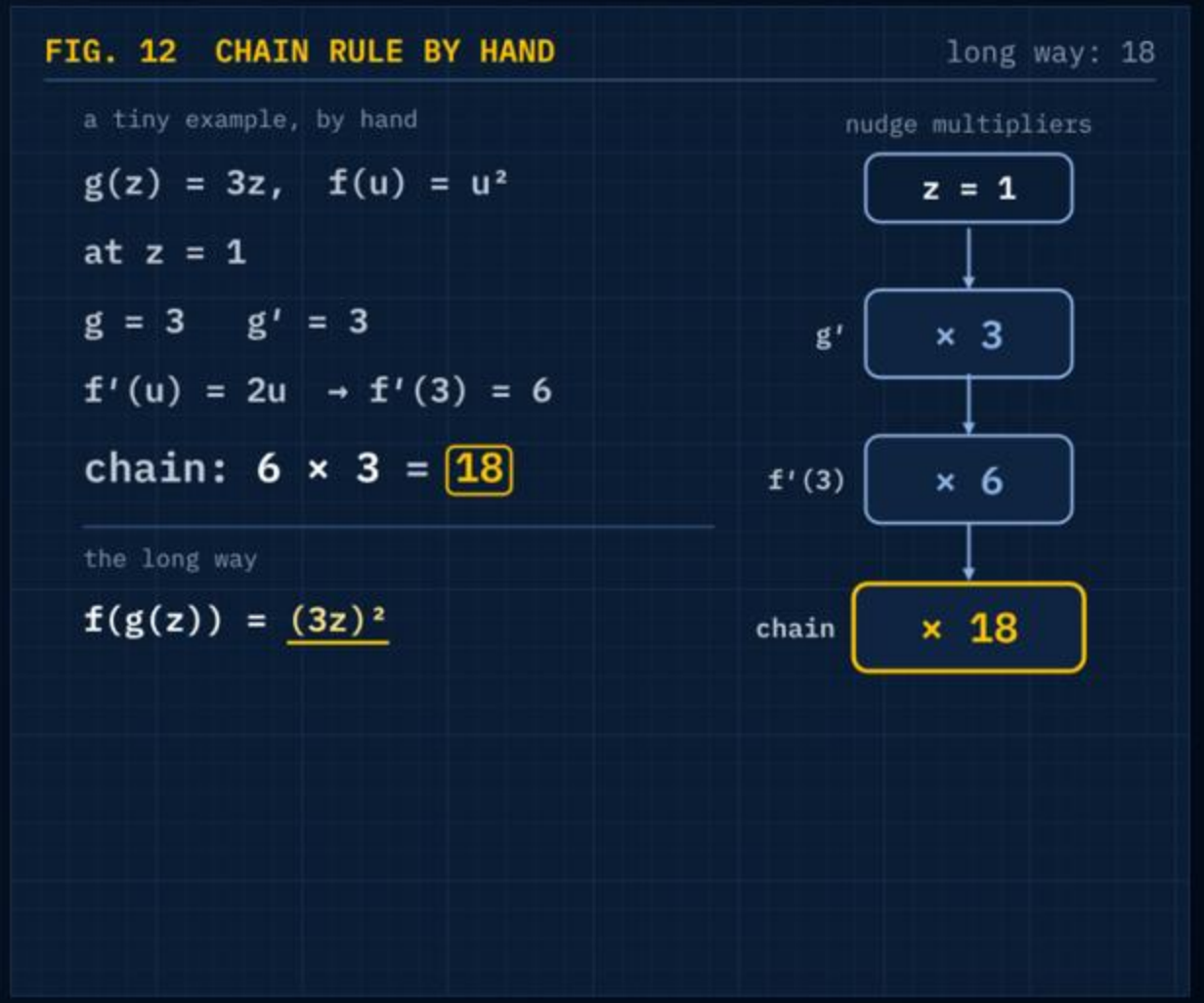
A tiny chain

$g(z) = 3z$, $f(u) = u^2$, at $z = 1$

01 $g = 3$, $g' = 3$, $f'(3) = 2 \times 3 = 6$

02 Chain rule: $6 \times 3 = 18$

03 Long way: $(3z)^2 = 9z^2 \rightarrow 18z \rightarrow 18$ ✓



Worked example; every number is checked in the companion notebook.

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PART 2 · THE REAL CHAIN

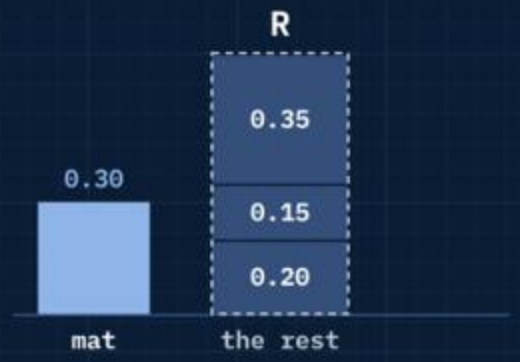
Score → probability → surprisal

z = mat's score; the other words lumped into R.

01 R = 0.20 + 0.15 + 0.35 = 0.70

02 p = g(z) = e^z ÷ (e^z + R) → 0.30 at z = ln 0.30

FIG. 13 THE SOFTMAX CHAIN R = 0.70



z = score for mat

R = 0.20 + 0.15 + 0.35
= 0.70

$p = g(z) = e^z \div (e^z + R)$

$z = \ln 0.30 \rightarrow e^z = 0.30$

g =

Softmax from Episode 4, with the other scores held fixed. Episode 1's probabilities (mat 0.30, floor 0.20, sofa 0.15, other 0.35) are illustrative, not measured.

MARK

MATH LAB · THE TWO LINKS

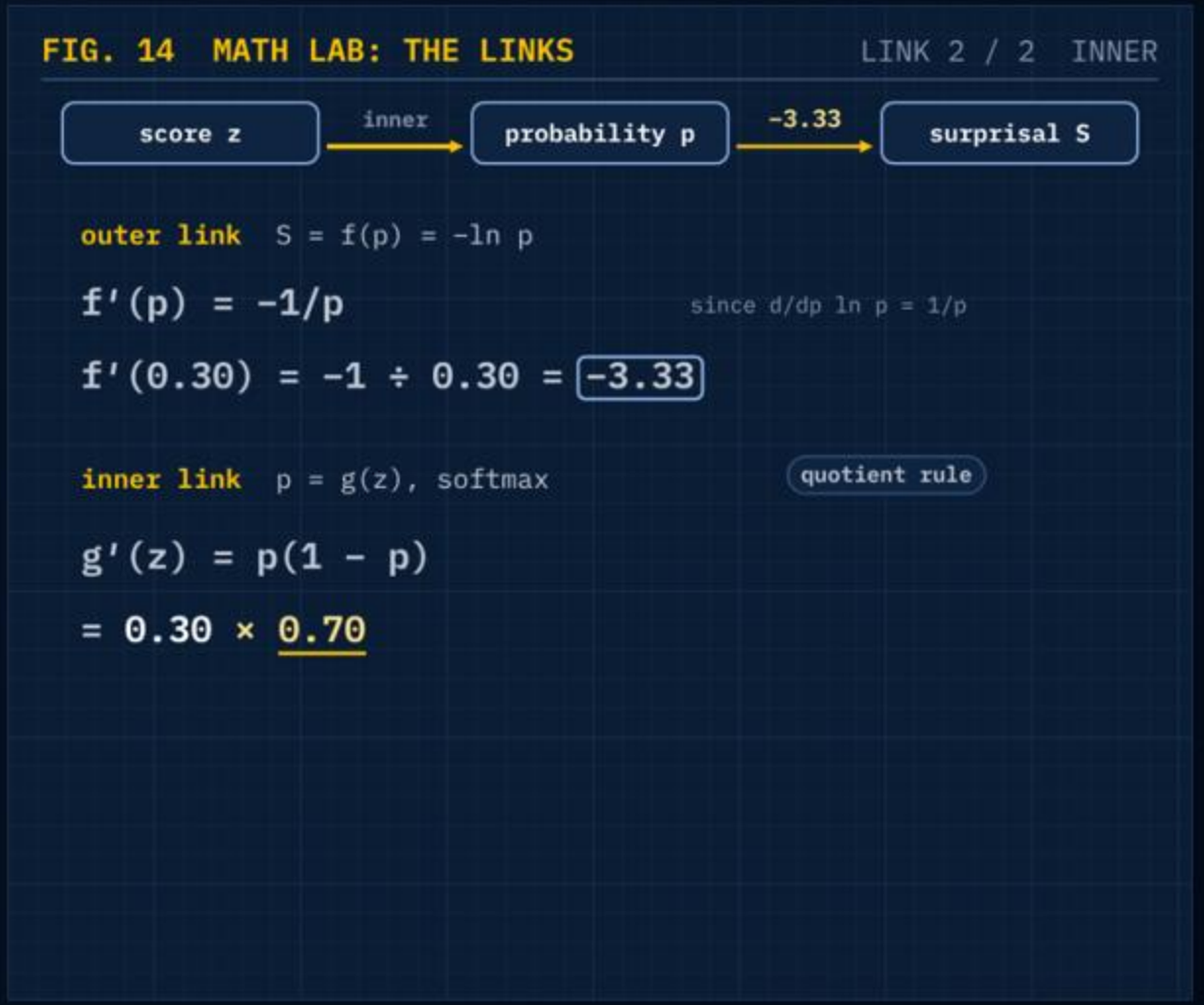
Outer link, inner link

Each link has its own slope.

01 Outer: f'(p) = −1/p = −3.33

02 Inner: g'(z) = p(1 − p) (quotient rule)

03 0.30 × 0.70 = 0.21



OpenStax, *Calculus Volume 1*, ch. 3 (differentiation rules, chain rule). Worked example; every number is checked in the companion notebook.

The quotient rule

p = e^z ÷ (e^z + R): a top over a bottom.

01 (top/bottom)' = (top'·bottom − top·bottom') ÷ bottom²

FIG. 15 QUOTIENT RULE — STEP 3 / 3 SLOPES

$$g'(z) = p(1 - p)$$ the step most explanations skip

$$p = \frac{e^z}{e^z + R}$$ ← top ← bottom

THE QUOTIENT RULE

$$(\text{top} / \text{bottom})' = \frac{\text{top}' \cdot \text{bottom} - \text{top} \cdot \text{bottom}'}{\text{bottom}^2}$$

two slopes:

MATH LAB · PLUG IN

Watch it cancel

e^z·(e^z + R) − e^z·e^z

01 The e^z·e^z pieces cancel: R·e^z is left

02 g' = R·e^z ÷ (e^z + R)²

03 Numbers: 0.70 × 0.30 ÷ 1² = 0.21

FIG. 16 WORKSHEET: PLUG IN — NUMBERS

numerator: top′·bottom − top·bottom′

$$e^z\cdot(e^z + R) - e^z\cdot e^z$$

$$= \cancel{e^z\cdot e^z} + R\cdot e^z - \cancel{e^z\cdot e^z}$$ multiply out

$$= R\cdot e^z$$ left on top

$$g'(z) = \frac{R\cdot e^z}{(e^z + R)^2}$$

the numbers, at z = ln 0.30

$$e^z = 0.30$$

Worked example; every number is checked in the companion notebook.

MATH LAB · THE NEAT PART

p(1 – p)

e^z ÷ bottom = p, and R ÷ bottom = 1 – p.

01 g' = p(1 – p) = 0.30 × 0.70 = 0.21

02 Same answer, two ways

03 The slope of a softmax probability with respect to its own score

FIG. 17 REWRITE: p(1 – p) TWO WAYS ✓

you'll see p(1 – p) again

$$g'(z) = \frac{R \cdot e^z}{(e^z + R)^2}$$

$$= \frac{e^z}{e^z + R} \cdot \frac{R}{e^z + R}$$

$= p$ $= 1 - p$ the two parts add to 1

$$g' = p(1 - p) = 0.30 \times 0.70 = \boxed{0.21}$$

slope of a softmax p w.r.t. its own score

route 1: quotient rule

$0.70 \times 0.30 \div 1^2$

$= 0.21$ ✓

route 2: rewrite: p(1 – p)

0.30×0.70

$= 0.21$ ✓

Worked example; every number is checked in the companion notebook.

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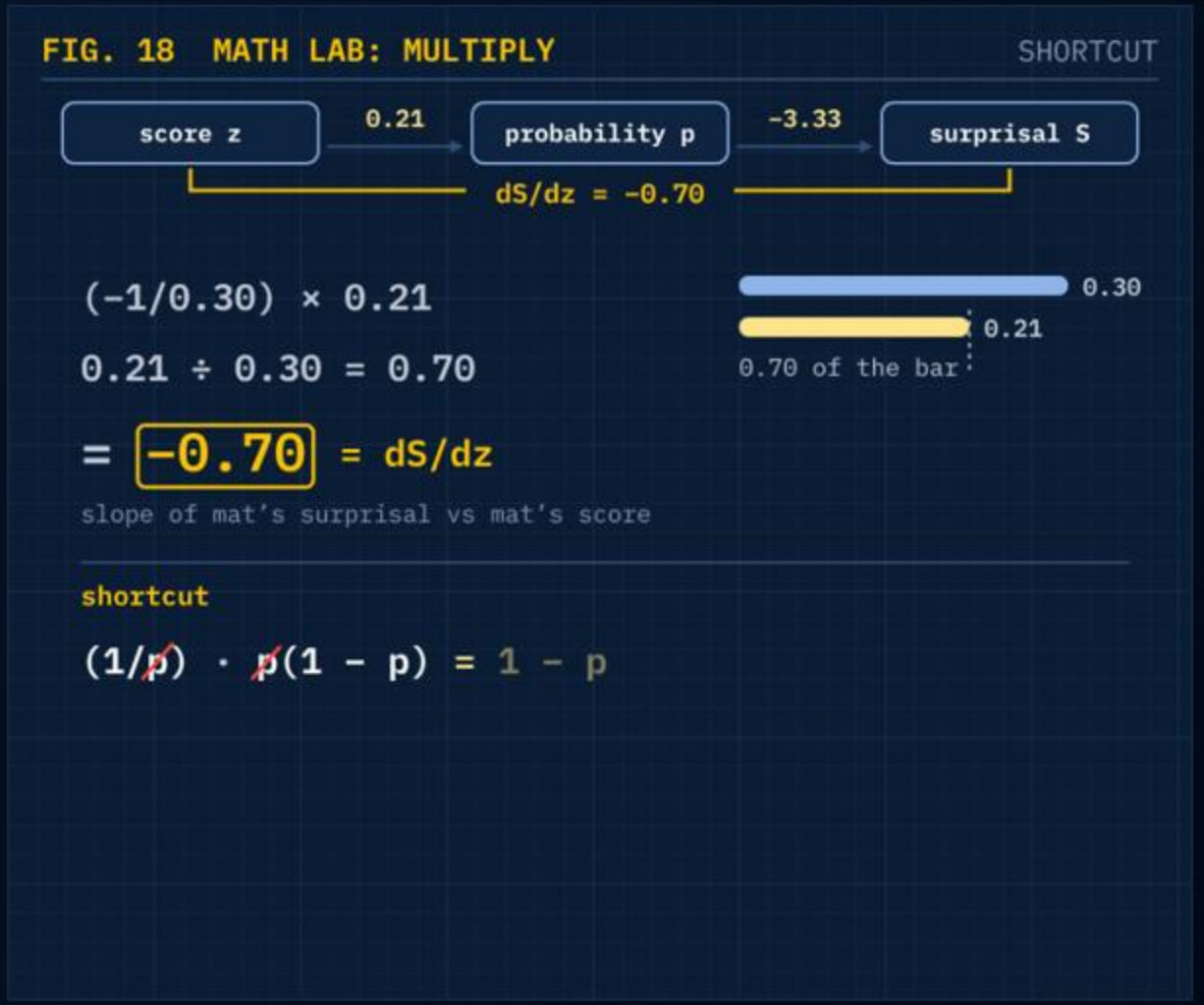
MATH LAB · MULTIPLY

dS/dz = −0.70

$(-1/0.30) \times 0.21 = -0.70$

01 $0.21 \div 0.30 = 0.70$

02 Shortcut: $(1/p) \cdot p(1-p) = 1-p$



Worked example; every number is checked in the companion notebook.

MARK

MATH LAB · TEST THE PREDICTION

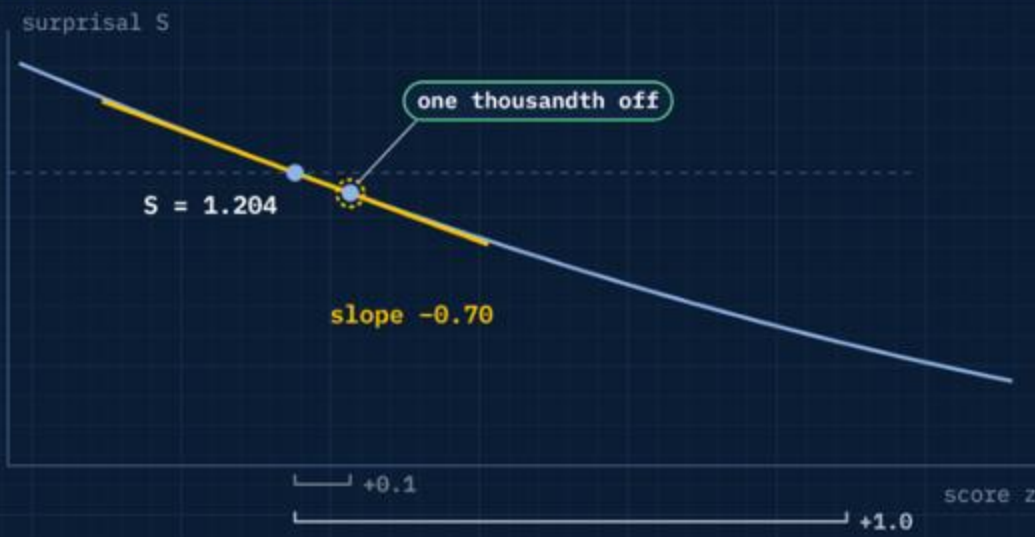
Small steps: spot on

Raise mat's score by 0.1, then by 1.0.

01 +0.1: predict −0.070; actual p 0.300 → 0.321, S 1.204 → 1.135 (−0.069)

02 +1.0: predict −0.70; actual −0.584

FIG. 19 TEST THE SLOPE Δz = 1.0



small nudge Δz = 0.1		big step Δz = 1.0
predict	0.1 × 0.70 = 0.070	
p	0.300 → 0.321	
S	1.204 → 1.135	
drop	0.069	

Worked example; every number is checked in the companion notebook.

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RECAP

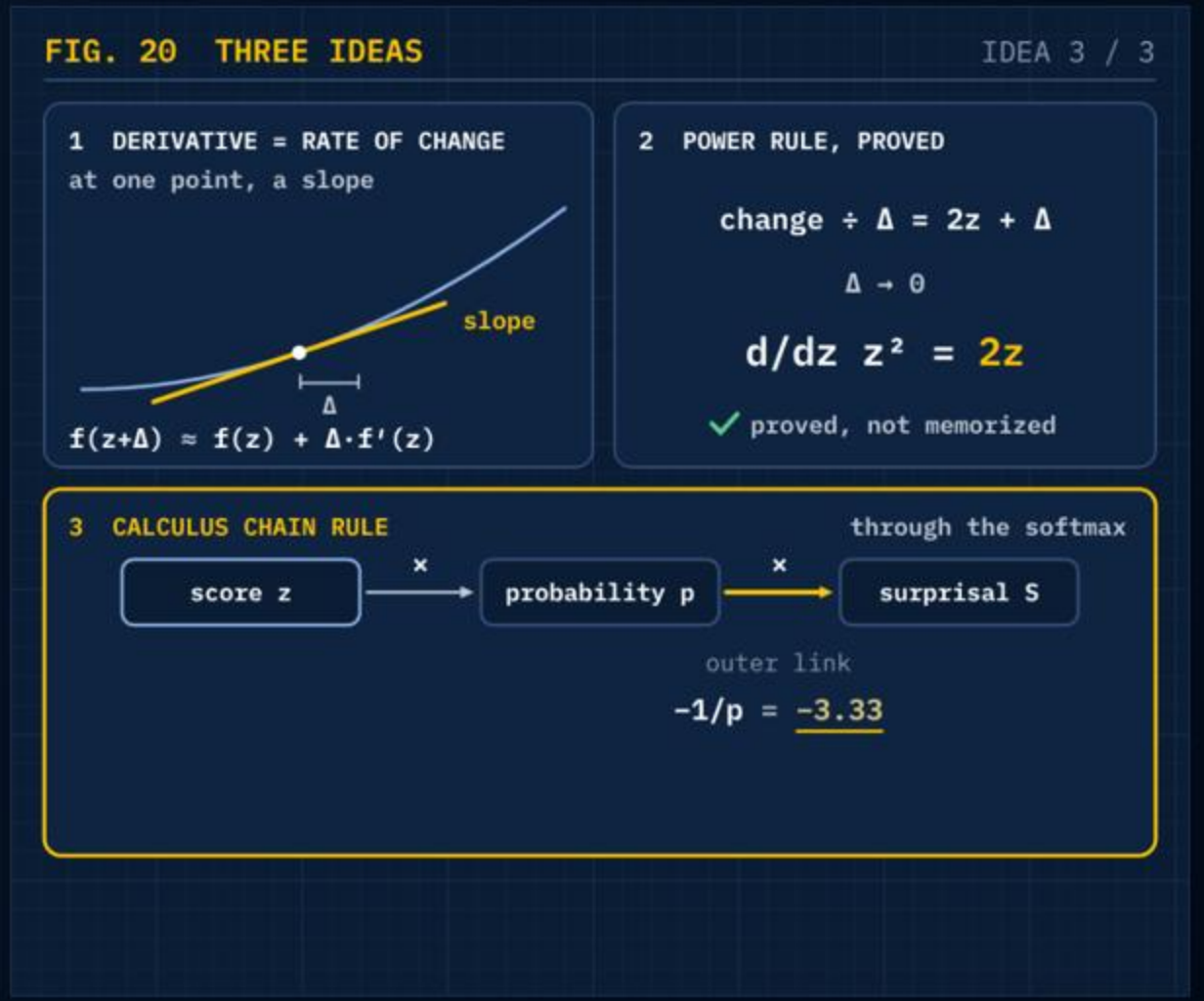
Three ideas

Part 2 builds on all three.

01 A derivative is a rate of change: $f(z + \Delta) \approx f(z) + \Delta \cdot f'(z)$

02 The power rule, proved: $2z + \Delta \to 2z$

03 Chain rule through softmax: $-3.33 \times 0.21 = -0.70$



All sources are listed in the description.

MARK

YOUR TURN

One question to take with you

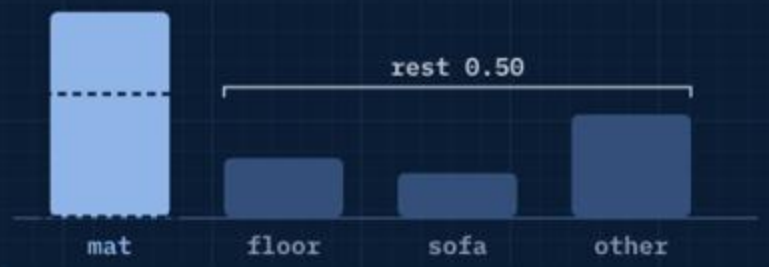
Suppose mat's probability were 0.50 instead of 0.30.

01 Slope of surprisal with respect to p?

02 The inner link, p(1 – p)?

03 Slope of surprisal with respect to mat's score?

FIG. 21 CHECK YOURSELF — QUESTION 3 / 3



mat's p

0.50

instead of 0.30

(a) dS/dp at p = 0.50 ? — surprisal vs p

(b) p(1 – p) ? — the inner link

(c)

Worked example; every number is checked in the companion notebook.

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YOUR TURN

The answer

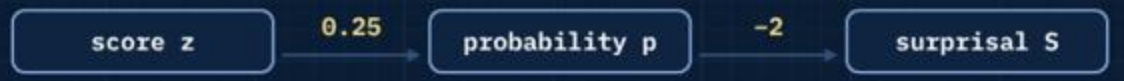
Push a nudge through a softmax by hand.

01 $-1 \div 0.50 = -2$

02 $0.50 \times 0.50 = 0.25$

03 $-2 \times 0.25 = -0.50 = -(1 - p)$

FIG. 22 THE ANSWER — SHORTCUT



(a) outer link: surprisal vs p

$$dS/dp = -1 \div 0.50 = -2$$

(b) inner link: p vs score

$$p(1 - p) = 0.50 \times 0.50 = 0.25$$

(c) multiply the links

$$dS/dz = -2 \times 0.25 = \boxed{-0.50}$$

shortcut: $(1/p) \cdot p(1 - p) = 1 - p$

$$dS/dz = -(1 - p)$$

Worked example; every number is checked in the companion notebook.

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TRY THIS WEEK

Three things to try.

01

Find the slope of z^3 at 2 with steps of 1, 0.1 and 0.01. Then prove the rule like we did for z^2

02

In the notebook, nudge mat's score by 0.1 and confirm the drop of about 0.07

03

Redo the quotient rule with p = 0.50, all on paper

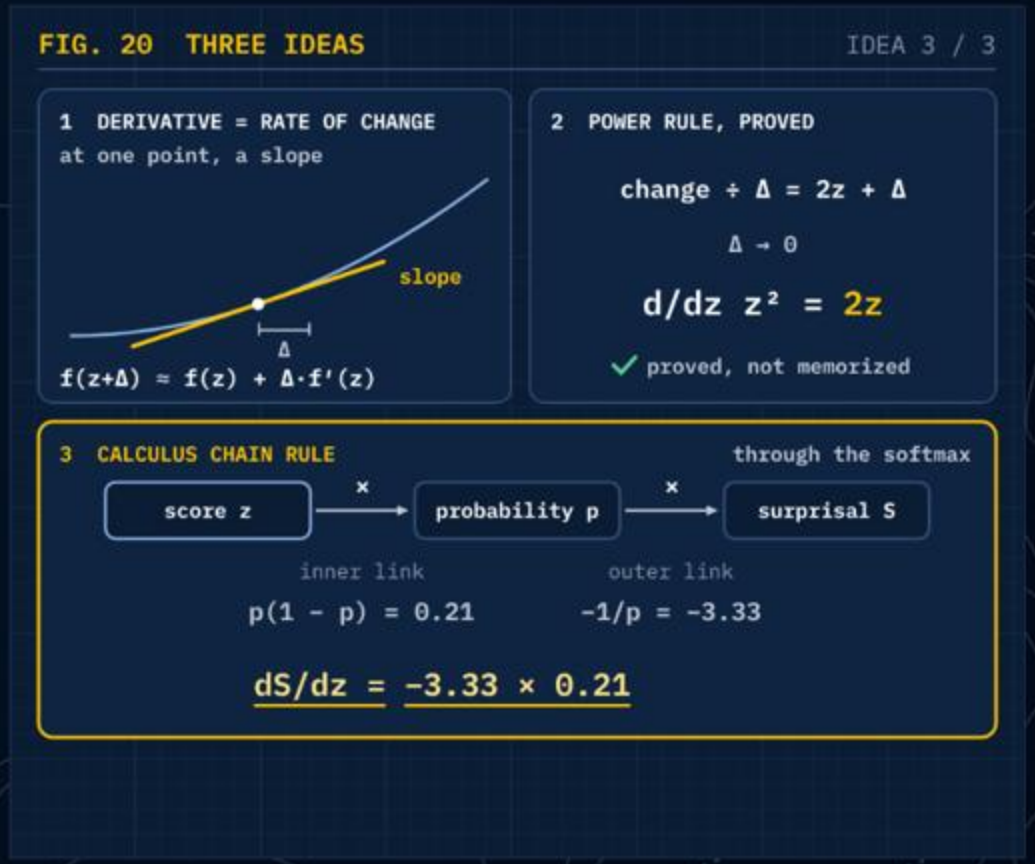
Companion notebook and every source in the description.

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NEXT TIME · PART 2

Episode 6: The gradient and the step.
One step downhill. Land at 19?



In the comments: which step was new to you, proving the power rule, or the quotient rule through softmax?

How Large Language Models Work · subscribe or follow for Part 2

MARK